

A Packing of 15 Four-Spaces in $\mathbb{R}^{12}$

Limin Ge*

October 9, 2026

Abstract

We give 15 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.856822. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.844075, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 15 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.856822\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 15$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.844075, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 15/14 = 2.857143$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N15.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
28568228172216963026373088592921297624067584079908055026397894615946852
248233445252714640727587089095472054648168045900653986305233671018369291
4965852966042878975344067876815241305858314941108
```

denominator:

```
99999999999162714171836501485868395616468118107452920608086039947156752
660021788280899483841821293657218831311973543577029390270870869530447240
422666442472954374602748476928463756517960716407
```

B Coordinates

15 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N15.txt`.

```
-414649310727 253933106182 102026850951 2191615256853684438530 190229307874 -365682441975 -468025022871625136539 426019651707 137988475126 -524136636738
-37303992800 -421312980550 -309659681494 315869590720243035432 -64305430625 213521114967 -2795561721485597815065 120356678954 -172003785848 -137449371806
-103707255918 -46268585581 -82513648838 14497633095260342479201 -357531283712 473729030481 -5808940883491425322228 -316285530307 643247626363 -127560440814
-181344790445 289733758913 259840159772 176920274774846245282724 141823644782 9368400050 -62605439363 -305776758006 142879269610 -366390210427 -153835689132
-263435254706 26081459621 334347448581 -288870780990897309104274 -92219742462 148094503613 14223279221550106686304 -500433697556 -558530614686 -207725889927
-15278819994 -532213965133 186183904753 -1517240331462561754220 266939999345 65077867662 36855282214 409049779736 299998392461 27974704513 378476097826
391654982139 209087005935 333514111722 -681603719725056200226 -330951491195 147474279133 60381589800 93349275498 -51702622268 -2011191565 -481957063874
-177155911044 -471579111676 607474456199 1051824760629211514258 -313676391361 207338132912 152823692281943078620 -6988875855 1383284449371 -188005517894
129021696605 76653273925 396152591994 199306191590 -323151209163 -412854086193 -469801759798 1836332493608497091 -228063691602 -131478254656 -43115117951
287071276972 209463227498 127626218458 -39336232499329511601653 -66479365691 -424816210216 -53847126628642826639 -455116934110 154105784147 232785420569
463590095924 49435849676 136266417093 659491635095 -323233364469 458719470627 44058881838 -10730795725237849248978 272847864037 -56970524950 161362928014
463304999296 -264000499603 -29539597924 -219804262336145793339 -372505422127 -319608444177 280731616822369488073 82614460663 -164942359534 361029808739
592137665509 166184034322 218596893567 -29851320057249199920041 135979157564 -164955505642 311982009330886590850 557536040917 1011099412 -114172264756
-189889024639 -439539833359 297992985483 7524686085181667879703 -275612810013 -87116887673 -123211965757423557862 234489919218 -181244848740 218050629904
334154628071 -235824843644 -517979342049 -324429890233144945154 228122926181 -274358361695 46420708898721681711142 -20405059668 7812376125 295607218564
-47005678724 -39759154714 -3203209064018 34028563568756980524678 662378006070 38072915192 70851805702 -3606861582 475716792753 341832113594 379561209487
211734664403 -265534360239 1949771517627 -30252829843354677594 357555764299 -107936892951 -46667552227018886146240 357138671379 -196244142442 -39370380886
28386992347 -201779370529 -41373799103 13353618699423898515699 259551372047 1047035920007 -2440453452672016519321 -277373381913 -86057437188 -180635383137
-101089469545 447740531058 -548810362795 -44243586353022948735 -354372506446 -175151940340 -137285587200932345888 108074519361 385435230197 -333172566114
112372293896 156060470157 134633301671 -98366681519284421763044 -268435362867 -258936972806 -2233181238052016310 -93871533923 -274191125245 195785253473
-517704719557 169850190958 203476567485 -25433294998127119494 -141012175680 628875609274 429892613520100488367924 -239382166658 158367191180 -291962278069
365928128607 405551269140 307878652873 46059756287292036668042 -26396253625 -31654048377 -220514568830211644886240 -199831805409 -222554252589 240123057156
14973433983 36367097278 90594203029 291303745406 463558096987 -35272400096 -186485811842 -1245996296272630253935 31167491404 -549437877803 182843055547
169754280352 -439288599588 -124111625736 457209012030818644049 -9314532117 -144939339351 348322005480303367521434 290277854140 -451760769278 -590017497342
327789407698 -290332740774 383281347584 -44772053463143303325771 258683603961 304585507087 -115550863680541213791 -149999199024 446070288618 -39647504600
283505197009 -403215381833 223281272193 -109852780871354353001 -300488600760 -87531116031 386221746281941557455 -733700087636 -108133766973 -149893547647
-196531310482 162673683498 -298292019265 -2632618730279439919708 -333937842835 -206705575682 303188464352928696281 -19096272529 -121120700315 4595748003388
-403652025113 -518390244139 -218606185727 191430345298000795865 -61224552128 404790536620 3950058368894388227204 84392330384 -49776767174 -144143756655
14917456406 18978209473 -361611002043 246856492631448753419876 27830866564 -53461765823 137304169330030403611 -107458924111 109105804865 682298223665
101058355388 -251533758107 -479646790434 -16403280588154037254 479128513811 63230628597 526428980506 209970001402 -40735490076 -253665733547 156578184625
139280800472 -397019366014 1652923351 235776061923 -57265398648 153811474385 -426321786035 46508503895818562003735 57546048832 192114503849 -276705367162
-48811580918 157959850976 40410089702 -149423510576173859767 2178391372 -541864417662 -1123591649160295912173 163053468274 -228799414363 314251648393
-298172674442 -405059285896 187279483525 -4649274630004946645 -161246096534 -1826054633 93920552281 -362679879371 -386926307717 210752284064 -1105131519090
-212516304681 89786171755 165093345124 357492954505455839019939 505286747846 -22383875573 -17062340228844492940 -368449404463 -265818681923 -174369254747
-412212055023 -194846352473 -45834755234 421362589233079928503 -329063415587 -118629856219 -31491837975972345825 431942136 700668118764 169103363791
128387443815 40352258332 279859001005 438984759760 -104323633047 -85124446576 379373565932 12361712914250019494236 -101285712261 -25709731126 92481369249
165916474120 368585940391 -120848007841 9641921036233630268811 30469781609 -250485079304 -30021278131292214970145 -110640681109 510392307078 -266298913921
-137494050323 267565627856 526327331510 33705028123724149733561 -217733527569 -259928815481 -37730494483480243630 -10867642839 -47332188832 -180369990982
-96984473744 272918506773 -231112789969 137101132876790345309 -355585187485 2230394633452 -24589112084868655849464 320790543420 185723439693 256950738556
-551157001114 -260882542417 33211939518 381301780128112933833 248594818388 -89881207841 162302010987 364105995780 -404371344443 383366420896 84697865601
111411954631 -286808499884 106271504105 -282736623040497094736 -230839664099 162470359849 335243612556198780789 194375211643 -349778271849 -170435264917
221199815510 260697665262 -220481458591 2006187414536935026156 278740848248 38689491269 -63280520727 86317385074 272005804306 59665980311 -710930263248
302801507049 -289430570342 444969702216 -278468546507360687964 344511746242 645035072247 -4853467236662847677638 -523401180530 214301482942 -107694795221
181233704742 -486543897296 133769879670 5863034758184675307292 97226622759 -242683669935 -5371310443277335655749 -344396380297 -298959765015 -163346567971
214496034527 -168052094785 -262912521630 27685431616183290257 -443624807483 290048706719 -12897655860765733451545 171820142146 295612362521 -52101391140
83247462900 288789209441 364108751974 184691926355 -240202657879 215105170837 -318991518765 -176657763288616697871 84386777271 212316781692 380087132716
-96616752336 250160117572 416716911637 -243354139035660316632 -507933227378 -186931480402 32391494097269060751046 404190595843 388918688084 -73058361387
-626288336939 -71208294887 -52065316401 -2780817744331699791984 -1084546212260 296955662117 -604561082585108785356 -129914998578 104241389559 521886123335
377064528826 92568017266 54888597577 232992297330 -162414601399 -13487610220 184498071388332658603 308445407065 -184607491754 5321865561678
-125550074135 -243679416287 -6750919063 -288608559382614063618 258042083413 136097681325 3258737948392607184193 39707772015 342206985295 -189040379343
23059659752 -90004598129 222599766788 -41278669931201287522079 1672244743278 -5284466957260 -1783852832322349830764 -282309844300 -197042008747 1839721211955
-10278946655 591501357054 -490088938568 -226673353826741181174 2000650423023 464510879278 233300430902155338749490 -310232635726 -339345748614 -139071532882
811646835905 -80003348849 -319915308542 -438177366987854037784 108843680971 188381363600 -34331221572-513062774770 566994407108 231601478076 -83338108773
-19888923334 81060816647 3260049647 -6148805816 111972685516 15165727582 149100561517 -5127606801186976532596 112811363268 -20864168627 510582416418
-203776485853 40589487981 246143357414 208873673101 98926957110 -97419600253 -380207397956 -83535157162-30878520660 -296014547758 199221785258 69754426789
131725079191 -30881504330 -168990384585 32717243825626898314 -514337866847 384059705625 -32719504203026433879 195342721215 -639670369336 -304571113759
-76246908174 -4289930445578 -687632148899 -105098715899139763816 360106695923 162382551649 -64253926463625308933294 -384989907002 -20057215959 135551033462
213731673748 386179601860 120028229278 -4582696908087737803355 45828838252 27006043042 -355762388613 -226174748461 -337326496409 39392423281 169527150791
67273394396 -431970146447 162112854225 -20822235233434241466426 -171420832719 202910041433 -237505629238541968177 -62806403291 -130773158651 -444489266319
137461738898 196815412284 -46666720154 553233008625 213913130106 -197901915074 99425402503 -378948365998070039068230 -37542462293 -172571417441 -166160471255
```